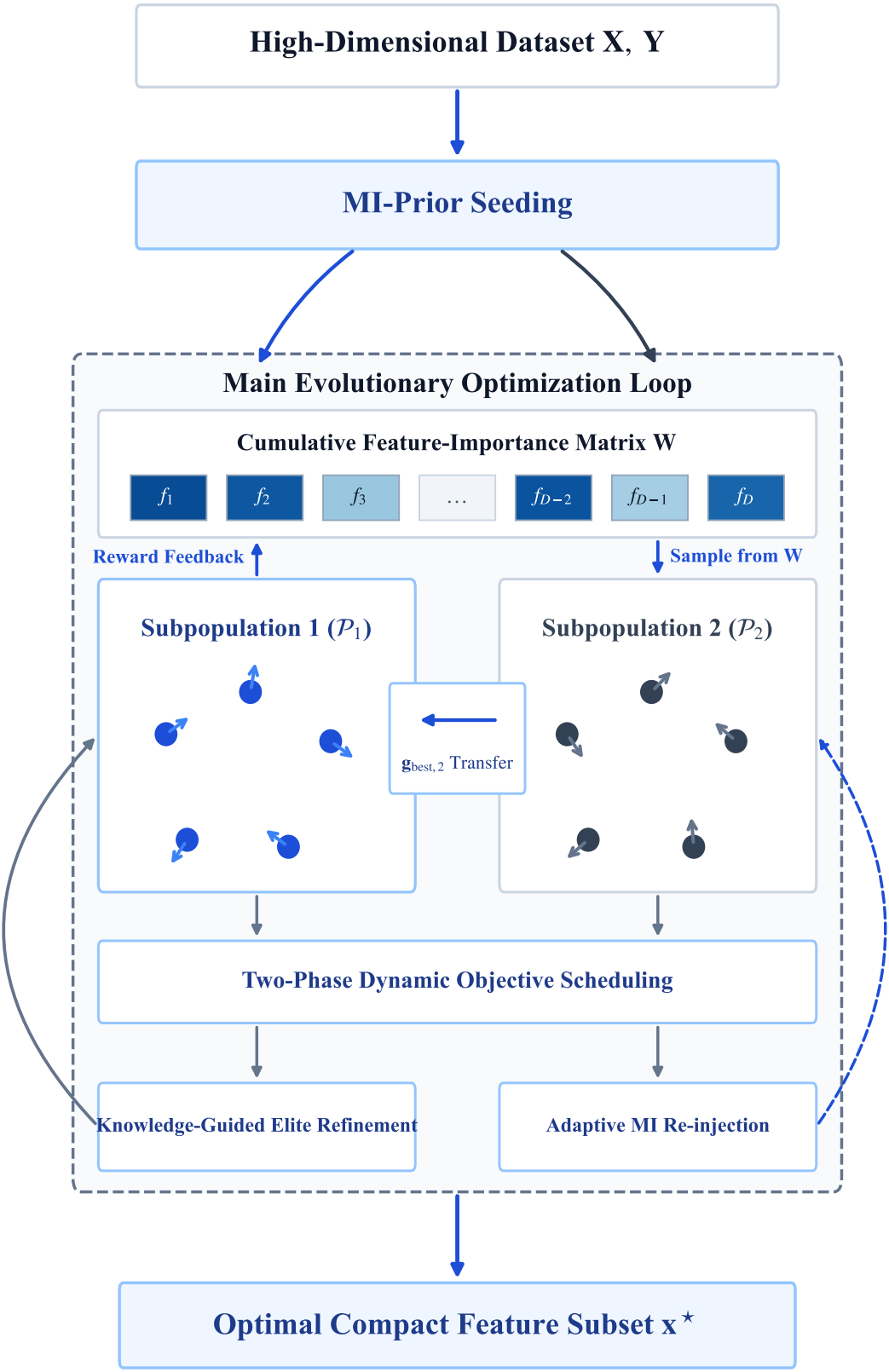
High-Dimensional Dataset X, Y
MI-Prior Seeding
Main Evolutionary Optimization Loop
Cumulative Feature-Importance Matrix W
f_1
f_2
f_3
…
f_{D-2}
f_{D-1}
f_D
Reward Feedback
Sample from W
Subpopulation 1 ($\mathcal{P}_1$)
Subpopulation 2 ($\mathcal{P}_2$)
$g_{best,2}$ Transfer
Two-Phase Dynamic Objective Scheduling
Knowledge-Guided Elite Refinement
Adaptive MI Re-injection
Optimal Compact Feature Subset $x^\star$